



FINAL REPORT

Challenge@PoliTo_by Firms – Simplify Hidden Scientific References

RefHunters

1. SECTION: DESCRIPTION OF THE PROBLEM

1.1. THE PROBLEM THAT YOUR SOLUTION WANTS TO SOLVE

Accessing reliable and up-to-date scientific knowledge is a critical requirement for students, researchers, and professionals. However, the current process of exploring academic literature is often slow, fragmented, and cognitively demanding. While a vast amount of scientific material is available through digital repositories, users still struggle to efficiently identify relevant information, verify its credibility, and understand how knowledge is connected across multiple papers.

Traditional search engines and academic platforms typically return lists of papers without offering clear, explainable links between answers and their original sources. As a result, users must manually read, cross-check, and interpret multiple documents to validate information, increasing the risk of misinterpretation, outdated references, or incomplete understanding. This issue becomes even more critical when dealing with complex or interdisciplinary research topics, where insights are distributed across citations and related works.

Moreover, existing AI-based tools often generate answers without sufficient transparency regarding where the information comes from, making it difficult to trust or reuse the output in academic or professional contexts. The lack of structured citation tracing, source validation, and thematic exploration limits their usefulness for scientific research and decision-making.

The problem, therefore, lies in the absence of a system that can reliably answer complex questions using only verified scientific sources, while also allowing users to explore citations, contextual relationships, and the evolution of knowledge in a transparent and structured way. Addressing this gap is essential to reduce research time, increase trust in AI-generated outputs, and support informed, evidence-based decisions in academia and industry.

1.2. HOW THE PROBLEM PROPOSED IS RESOLVED CURRENTLY

At present, the problem of accessing and validating scientific knowledge is addressed through a combination of traditional academic platforms and emerging AI-based tools. Digital repositories such as academic search engines and publisher databases provide access to large collections of



scientific papers, enabling users to search by keywords, authors, or citations. While these tools are effective in retrieving relevant documents, they largely rely on manual exploration: users must read multiple papers, identify relevant sections, and independently verify sources to build a complete and reliable understanding.

More recently, AI-powered research assistants and language models have been introduced to support literature analysis by generating summaries or answering questions based on scientific texts. These tools partially reduce the time required for information retrieval; however, they often lack sufficient transparency. In many cases, answers are generated without clearly indicating the exact sources, sections, or citations used, making it difficult to assess the reliability, correctness, and completeness of the output. This limitation significantly reduces their applicability in academic, professional, and regulated contexts.

Additionally, existing solutions typically do not support structured exploration of citations, cross-paper comparisons, or the analysis of how ideas evolve across related works. As a result, users still face fragmented workflows, switching between multiple tools to retrieve documents, validate sources, and interpret relationships between papers. Overall, current approaches provide access to information but do not fully address the need for trustworthy, explainable, and structured scientific knowledge exploration.

3. SECTION: SOLUTION PROPOSED BY THE TEAM

3.1.SHORT DESCRIPTION OF THE SOLUTION (max 1000 characters)

The proposed solution is a multi-agent generative AI system designed to answer complex user queries using only verified scientific sources. Starting from a publicly accessible academic paper, the system retrieves, analyses, and cross-references its citations to generate reliable, evidence-based answers.

Each answer is transparently linked to specific passages in the original paper or its referenced works, allowing users to verify sources and explore the underlying scientific context. Multiple specialised agents collaborate to handle document parsing, citation extraction, reasoning, and answer generation, ensuring both accuracy and depth.

The system also enables structured exploration of related literature by highlighting thematic connections between papers and indicating the credibility and recency of sources. By combining explainable AI, citation traceability, and interactive exploration, the solution significantly reduces research time while increasing trust and usability in academic and professional settings.

3.2.TYPE OF SOLUTION



A product/device ☐

A system ☐

A processing method ☐

An algorithm ☐

Other ☐ (explain)

Comment and short motivation: The proposed solution is best classified as a **system**, as it integrates multiple specialised AI agents, external scientific data sources, and user-facing interfaces into a single coherent architecture. Rather than focusing on a standalone algorithm or isolated processing method, the solution orchestrates document parsing, citation retrieval, reasoning, and answer generation as interconnected components. This systemic approach enables transparent, reliable, and explainable exploration of scientific knowledge, aligning with real-world academic and professional use cases.

3.3.APPLICATION AREAS OF THE SOLUTION

The proposed solution is designed to support organisations and institutions that rely on **accurate, verifiable, and up-to-date scientific knowledge** for research, decision-making, and compliance. RefHunters addresses the needs of multiple application areas where transparency, traceability, and efficient access to scientific information are essential.

In **academic institutions and research centres**, the system supports literature review activities by enabling structured exploration of scientific papers, citation analysis, and contextual understanding of research contributions. By providing transparent, citation-backed answers and adaptive explanations based on expertise level, RefHunters improves research efficiency, reduces manual effort, and supports higher-quality academic outputs for students, researchers, and educators.

In **corporate R&D and innovation units**, the solution facilitates faster and more reliable assessment of scientific evidence. By enabling direct comparison of methods and results across papers, RefHunters supports monitoring of emerging technologies and trends, contributing to informed strategic decision-making. This is particularly valuable in innovation-driven environments where time-to-insight and evidence-based evaluation are critical.

The system is also highly relevant for **consulting and advisory services**, where credibility, source validation, and traceability are fundamental requirements. RefHunters enables consultants to produce analyses and reports supported by clearly referenced scientific evidence, reducing the risk of misinterpretation and increasing confidence in AI-assisted recommendations.

Furthermore, RefHunters is applicable to **knowledge-intensive and regulated industries**, such as healthcare, engineering, and high-technology sectors. In these contexts, access to validated scientific information is crucial for product design, development processes, risk assessment, and regulatory compliance. The system's transparency and explainability features help ensure that decisions are grounded in reliable and auditable scientific sources.



Finally, the solution can be integrated into **digital knowledge management platforms** to enhance information retrieval and credibility assessment. By embedding AI-assisted, citation-aware reasoning into existing systems, organisations can improve the explainability and trustworthiness of knowledge-intensive workflows, supporting long-term knowledge management and decision support.

3.4. ACCURATE DESCRIPTION OF THE SOLUTION

RefHunters is a web-based research assistant that allows a user to upload a scientific PDF (or provide a DOI), ask a question, and receive an answer that is grounded in **verifiable evidence** and linked to **exact locations** inside the document.

The system integrates four operational capabilities:

1. Document ingestion and structuring

- The backend receives a PDF and processes it in two complementary ways:
 - extracting structured document metadata and citations via a local **GROBID** service,
 - tracking raw text locations so that passages can be highlighted precisely in the PDF viewer.
- Citations found in the bibliography are normalized into a registry that the system can use to fetch referenced papers when needed.

2. Evidence collection from verified scientific sources

- When the user query requires external context, the system searches and downloads related papers from **arXiv** and adds them to the evidence set used to answer the question.

3. Answer generation with traceability

- Answers are generated using **OpenAI Tools** and are explicitly tied to the evidence set; the UI exposes citations that map back to exact locations in the main PDF (and, when applicable, referenced papers).

4. Interactive refinement and expertise adaptation

- The user can control the explanation depth via expertise modes (beginner/intermediate/expert) and iteratively refine outputs through follow-up questions and feedback.

Implementation indications

Repo address: <https://github.com/Hesam-AHT/Agentic-Research-Assistant/tree/a0-openAI>

Final version of code is in branch: a0-openAI

A) Required Component (local development)



Frontend Web app(React) + Backend Api(Nodejs/ Typescript)+External Services (Redis + GROBID) +OpenAI API Key

B) Environment configuration

- Backend environment variables include OpenAI_API_KEY, backend PORT, and GROBID_URL.
- Frontend environment points to the backend via VITE_API_URL.

C) Start-up procedure (reproducible commands)

Start Redis and GROBID

- `redis-server --daemonize yes`
- `docker run -d --name grobid -p 8070:8070 grobid/grobid:0.8.0 OPERATIONS_GUIDE`

Start backend and frontend in separate terminals

- `cd Second-backend && npm run dev`
- `cd Second-Frontend && npm run dev`

3.4.1. Final Prototype

Below are the screenshots of the final prototype. You can see how the expertise level can be selected for each individual question. Based on the user's feedback, there are also three different ways to regenerate the answer.

The user can review the selected expertise level, and if it is set to Beginner, the system tries to answer using only the **main paper** (not the references). You can see this in the last two screenshots: in the first, only the main paper is used even though the question is comparative; in the latter, when the expertise is Expert the system also extracts evidence from the references.



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http://localhost:5174

Sign in

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Simplify Hidden Scientific References

Upload PDF



Click to upload or drag and drop
PDF files only

Expertise Level

Intermediate

Beginner

Intermediate

Expert

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http://localhost:5174/answer

80%

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RefHunters

New Chat

10 pages

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Progressive Semantic Segmentation

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Abstract

The objective of this work is to segment high-resolution images without overloading GPU memory usage or losing the fine details in the output segmentation map. The memory constraint means that we must either downsample the big image or divide the image into local patches for separate processing. However, the former approach would lose the fine details, while the latter can be ambiguous due to the lack of a global picture. In this work, we present MagNet, a multi-scale framework that resolves local ambiguity by looking at the image at multiple magnification levels. MagNet has multiple processing stages, where each stage corresponds to a magnification level, and the output of one stage is fed into the next stage for coarse-to-fine information propagation. Each stage analyzes the image at a higher resolution than the previous stage, recovering the previously lost details due to the lossy downsampling step, and the segmentation output is progressively refined through the processing stages. Experiments on three high-resolution datasets of urban views, aerial scenes, and medical images show that MagNet consistently outperforms the state-of-the-art methods by a significant margin. Code is available at <https://github.com/VinAIResearch/MagNet>.

Figure 1: Comparing several semantic segmentation and refinement approaches on a high-resolution input image. Downsampling loses fine details, while Patch Processing wrongly classifies local patches due to the lack of the global context. The collaborative global-local network GLNet fails due to the large discrepancy between the global and local branches. Post-processing and refinement methods such as DenseCRF and PixelRef can only correct small mistakes due to local inconsistency. MagNet outperforms other methods, thanks to a novel multi-scale segmentation and refinement framework. Best viewed in color.

1. Introduction

The current state-of-the-art (SOTA) semantic image segmentation techniques [1, 4, 16, 19, 21, 23, 26] are based on deep learning, where a convolutional neural network (CNN) takes an input image and outputs a segmentation map. Most of the existing techniques, however, assume that the entire segmentation process can be performed with a single feed-forward pass of the input image and the entire process can be fitted into GPU memory. Unfortunately, most existing techniques cannot handle high-resolution input images due to memory and other computational constraints. One approach to handle a large input image is to downsample it, but this results in a low-resolution segmentation map, which is not adequate for applications that require high-resolution

The paper titled "Progressive Semantic Segmentation" by Chuong Huynh et al. focuses on improving the segmentation of high-resolution images while managing GPU memory usage effectively. The main challenge is that processing large images can either lead to loss of fine details if downsampled or create ambiguities when divided into smaller patches. To address this, the authors propose a framework called MagNet, which uses a multi-scale approach to analyze images at different magnification levels. This allows for a progressive refinement of segmentation results, where each stage builds upon the previous one, recovering lost details and improving accuracy. Experiments demonstrate that MagNet outperforms existing state-of-the-art methods across various high-resolution datasets, including urban views and medical images. The code for this framework is available online for further exploration and use.

References (1)

[1] Huynh Progressive Semantic Segmentation CVPR 2021 paper ()

Main Paper • Section: Full Document

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Figure 1: Comparing several semantic segmentation and refinement approaches on a high-resolution input image. Downsampling loses fine details, while Patch Processing wrongly classifies local patches due to the lack of the global context. The collaborative global-local network GLNet fails due to the large discrepancy between the global and local branches. Post-processing and refinement methods such as DeepCRF and PointRead can only correct small mistakes due to local inconsistency. MagNet outperforms other methods, thanks to a novel multi-scale segmentation and refinement framework. Best viewed in color.

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The paper "Progressive Semantic Segmentation" by Huynh et al. introduces a novel framework called MagNet designed to tackle the challenges of segmenting high-resolution images while managing GPU memory constraints. Traditional methods often face limitations when processing large images, either by downsampling, which sacrifices detail, or by segmenting small patches independently, which can lead to ambiguity due to the lack of global context. MagNet addresses these issues by employing a multi-scale approach that analyzes images at various magnification levels, progressively refining segmentation outputs through multiple processing stages.

Key Features of MagNet:

- Multi-Scale Processing:** MagNet operates through a series of stages, each corresponding to a different scale. This allows the model to capture both global and local information effectively, refining the segmentation output at each stage.
- Refinement Module:** At each processing stage, a refinement module uses the output from the previous stage and the current scale's segmentation map to enhance accuracy, particularly in uncertain areas. This dual input helps in correcting mistakes made in earlier stages.
- Memory Efficiency:** By dividing the input image into manageable patches and processing them at various scales, MagNet avoids overwhelming GPU memory while still producing high-resolution outputs. This is crucial for applications requiring detailed segmentation, such as medical imaging or urban scene analysis.

Performance: The authors conducted experiments on three high-resolution datasets: Cityscapes, DeepGlobe, and Gleason. Results showed that MagNet consistently outperformed state-of-the-art methods, achieving significant improvements in mean Intersection over Union (mIoU) metrics, which measure the accuracy of segmentation.

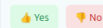
Example Application: For instance, in medical imaging, where detecting small malignant lesions is critical, MagNet's ability to maintain detail while processing large images ensures that important features are not lost. The framework's design allows it to refine segmentation outputs progressively, leading to more accurate diagnoses.

In conclusion, MagNet represents a significant advancement in semantic segmentation, particularly for high-resolution images. Its innovative multi-scale approach and efficient memory usage make it a valuable tool for various applications requiring precise image analysis.

References (1)

- [1] Huynh Progressive Semantic Segmentation CVPR 2021 paper (1)
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Figure 1: Comparing several semantic segmentation and refinement approaches on a high-resolution input image. Downsampling loses fine details, while Patch Processing wrongly classifies local patches due to the lack of the global context. The collaborative global-local network GLNet fails due to the large discrepancy between the global and local branches. Post-processing and refinement methods such as DeepCRF and PointRead can only correct small mistakes due to local inconsistency. MagNet outperforms other methods, thanks to a novel multi-scale segmentation and refinement framework. Best viewed in color.

output with fine details [12, 24], e.g., for tracking the progression of malignant lesion [8]. Another approach to handle a large input image is to divide the image into small patches and process each patch independently. This approach, however, does not take into account the global information [22] that is needed to resolve ambiguity in local patches. The limitations of these two approaches are illustrated in Fig. 1(c) & (d).

One way to address the limitations of the two approaches.

MagNet, SegFix, and DeepLab are all frameworks used for semantic segmentation, but they differ in their approaches and effectiveness.

MagNet

Architecture: MagNet is a multi-scale framework designed for ultra-high resolution images. It processes images in multiple stages, analyzing them at different scales to recover lost details and refine segmentation outputs progressively. Each stage uses a segmentation module and a refinement module to improve accuracy by addressing uncertainties in the segmentation maps [1][2].

Performance: Experiments show that MagNet consistently outperforms existing methods by 1% to 2% in mean Intersection over Union (mIoU) across various high-resolution datasets [3].

SegFix

Approach: SegFix enhances segmentation outputs by refining only the most uncertain pixels in the segmentation map. This selective refinement helps improve accuracy without requiring extensive computational resources [4].

Efficiency: It is designed to be efficient, targeting specific areas of uncertainty rather than processing the entire image at once, which can save time and resources.

DeepLab

Mechanism: DeepLab employs dilated convolutions and Atrous Spatial Pyramid Pooling (ASPP) to increase the receptive field and capture multi-scale contextual information effectively. This allows it to connect distant pixels and improve segmentation quality [4].

Use Cases: It is widely used in various applications due to its ability to handle different scales effectively, making it suitable for complex segmentation tasks.

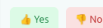
Summary

MagNet excels in high-resolution segmentation with a multi-stage approach, SegFix focuses on refining uncertain pixels for efficiency, and DeepLab uses advanced convolution techniques to enhance contextual understanding. Each method has its strengths depending on the specific requirements of the segmentation task.

References (4)

- [1] Huynh Progressive Semantic Segmentation CVPR 2021 paper (1)
[Main Paper](#) - [Section: Abstract](#)
- [2] Huynh Progressive Semantic Segmentation CVPR 2021 paper (1)
[Main Paper](#) - [Section: MagNet](#)
- [3] Huynh Progressive Semantic Segmentation CVPR 2021 paper (1)
[Main Paper](#) - [Section: Conclusions](#)
- [4] Huynh Progressive Semantic Segmentation CVPR 2021 paper (1)
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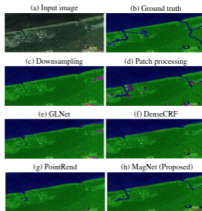


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One way to address the limitations of the two aforementioned

Comparison of MagNet, SegFix, and DeepLab

MagNet is a multi-scale segmentation framework designed for high-resolution images. It operates through a multi-stage architecture that processes images at various scales, refining segmentation outputs progressively. Each stage utilizes a segmentation module and a refinement module, which helps in recovering lost details from downsampling and resolving local ambiguities by integrating global context [1][2][3]. MagNet has demonstrated superior performance over existing methods, achieving a mean Intersection over Union (mIoU) improvement of 1-2% on various datasets [4][5].

DeepLab, on the other hand, employs atrous convolution and atrous spatial pyramid pooling (ASPP) to capture multi-scale contextual information effectively. This method allows for better localization of object boundaries by combining outputs from multiple layers and integrating a fully connected Conditional Random Field (CRF) for final segmentation refinement. DeepLab has set state-of-the-art results on several benchmarks, including PASCAL VOC and Cityscapes, achieving mIoU scores as high as 79.7% [6].

SegFix is a model-agnostic post-processing technique that enhances boundary quality in segmentation outputs from any model. It focuses on replacing unreliable boundary pixel predictions with those from more reliable interior pixels, thus improving segmentation accuracy without requiring retraining of the original model [7]. SegFix has shown consistent performance improvements across various segmentation models and datasets, ranking highly in competitions [7].

Key Differences:

Architecture: MagNet uses a multi-stage approach, while DeepLab utilizes atrous convolutions and CRFs. SegFix acts as a post-processing step applicable to any segmentation model.

Performance: MagNet outperforms many state-of-the-art methods, DeepLab achieves high mIoU scores, and SegFix enhances existing models without retraining.

Use Cases: MagNet is tailored for high-resolution images, DeepLab excels in dense prediction tasks, and SegFix is versatile across different models and datasets.

In summary, while MagNet and DeepLab focus on the segmentation process itself, SegFix enhances the output of existing segmentation models, making them complementary in practical applications.

References (7)

- [1] Huynh Progressive Semantic Segmentation CVPR 2021 paper (0)
[Main Paper](#) · [Section: Abstract](#)
- [2] Huynh Progressive Semantic Segmentation CVPR 2021 paper (0)
[Main Paper](#) · [Section: Introduction](#)
- [3] Huynh Progressive Semantic Segmentation CVPR 2021 paper (0)
[Main Paper](#) · [Section: MagNet](#)
- [4] Huynh Progressive Semantic Segmentation CVPR 2021 paper (0)
[Main Paper](#) · [Section: Experiments](#)
- [5] Huynh Progressive Semantic Segmentation CVPR 2021 paper (0)
[Main Paper](#) · [Section: Conclusions](#)
- [6] DeepLab: Semantic Image Segmentation with Deep Convolutional Nets, Atrous Convolution, and Fully Connected CRFs (2018)
- [7] SegFix: Model-Agnostic Boundary Refinement for Segmentation (2020)

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Expertise: Expert

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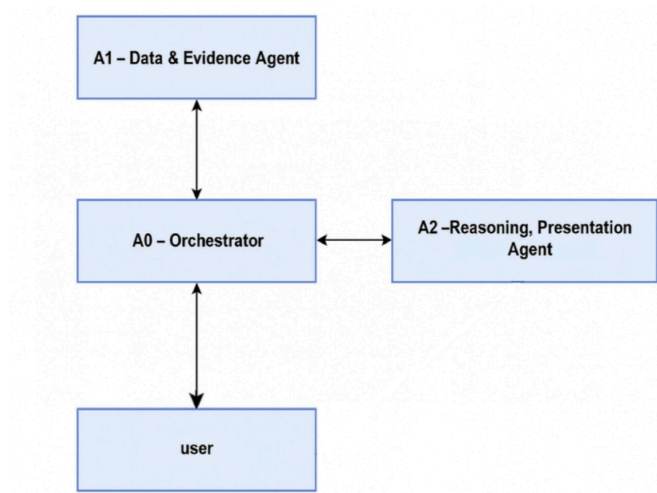
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3.4.2. System Architecture

The system is designed to function as a policy-aware, evidence-grounded research assistant. To achieve reliability, transparency, and extensibility, the architecture follows an agent-based modular design in which responsibilities are clearly separated and centrally coordinated.

At the highest level, the architecture is organized around a controller-worker paradigm, where a single controller agent (A0) governs the behavior of specialized worker agents (A1 and A2). This design ensures that reasoning and retrieval are never performed in an uncontrolled or implicit manner.



Architectural Principles

The system architecture is guided by the following principles:

1. **Separation of Concerns**

Control, retrieval, reasoning, and memory management are handled by different components.

2. **Centralized Policy Enforcement**

All decisions about what the system is allowed to do are made in one place (A0).

3. **Evidence-Grounded Reasoning**

Answers are generated only from explicitly provided evidence.

4. **Traceability and Transparency**

Every decision and output can be traced back to inputs and evidence.

5. **Extensibility**

New agents and capabilities can be added without redesigning the system.

3.4.2.1. Agent A0

A0 functions as the central control unit of the system. It is responsible for interpreting user input and deciding how the system should behave for each request. The user provides a query together with a paper reference (either a PDF file, Latex or a DOI), which A0 treats as the main context for the task.

The key responsibilities of A0 include:

- Interpreting user intent and query scope



- Performing policy planning to determine whether reasoning should be limited to the main paper or extended to external references
- Coordinating the execution of downstream agents
- Managing access to global memory
- Enforcing constraints to ensure policy-compliant behavior

A0 uses a lightweight language model exclusively for decision-making and planning, not for generating answers. This ensures that all system actions are explicitly controlled and traceable.

3.4.2.2. Agent A1

Agent A1 functions as the **Evidence Builder** and **Knowledge Retrieval Specialist** of the system. Its primary responsibility is to ingest the user's uploaded PDF, extract its semantic structure, and autonomously expand the knowledge base by retrieving external references. Unlike a standard RAG (Retrieval-Augmented Generation) pipeline that simply chunks text, Agent A1 builds a structured, state-aware "Evidence Object" that tracks the precise physical location of every claim.

The agent is implemented in TypeScript and utilizes a centralized **AgentState** class to maintain context during the ingestion and retrieval loops. The state persists three critical data structures:

1. **Main Paper Map:** A coordinate-aware mapping of the user's PDF, tracking text down to specific paragraph, line, and character offsets (**ExtractedTextWithLocations**).
2. **Citation Registry:** A structured list of references parsed from the bibliography.
3. **Evidence Collection:** A unified array containing content from both the main paper (segmented by section) and external downloaded papers.

Key Technical Capabilities

1. Hybrid Document Parsing Strategy

Agent A1 employs a dual-parsing approach to maximize accuracy:

- **GROBID Integration:** The agent communicates with a local GROBID server to extract high-level semantic structures (Title, Abstract, Introduction, Methods, etc.) and structured XML citation metadata.
- **Raw Text Location Tracking:** A custom **pdf-parse** implementation runs in parallel to map the text physically. This allows the agent to return **TextLocation** objects (e.g., "Paragraph 4, Line 12"), enabling the frontend to highlight exact sentences in the PDF viewer.

2. Robust Citation Extraction (Hybrid Regex/LLM)

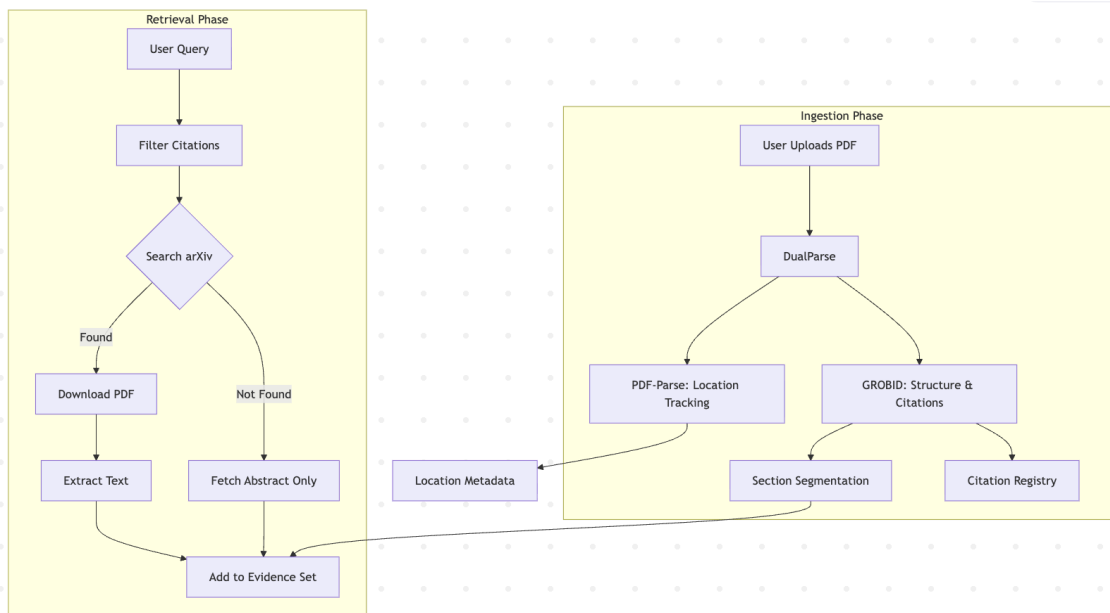
To ensure high fidelity in reference tracking, the agent uses a fallback mechanism:

1. **Direct XML Parsing:** It first attempts to parse GROBID XML output using strict Regex patterns to extract Title, Author, and DOI. This is fast and hallucination-free.
2. **LLM Fallback:** If direct parsing fails, the agent delegates the complex XML chunks to `gpt-4o-mini` to reconstruct the citation JSON.

3. Autonomous Retrieval Loop

When a user asks a question requiring external context, Agent A1 enters a retrieval loop:

- **Filtering:** It filters the internal citation registry based on keywords from the query.
- **arXiv Interaction:** It uses a rate-limited fetcher (with exponential backoff) to search for the papers on arXiv.
- **Ingestion:** Upon finding a paper, it downloads the PDF, reads the text, truncates it to a manageable context window (default 20k characters), and adds it to the global Evidence set.



3.4.2.3. Agent A2

Agent A2 acts as the **Analytical Engine** of the system. While the previous agent (A1) focuses on finding documents, Agent A2 is strictly responsible for reading that information, reasoning about it, and writing the final answer. It operates on a "Read-Analyze-Write" loop, ensuring that every sentence it generates is strictly grounded in the evidence provided.



The agent uses a persistent **AgentState** to manage the synthesis lifecycle. Key components include:

- **Evidence Buffer:** A read-only array of **Evidence** chunks (from PDFs and arXiv abstracts).
- **Expertise Controller:** A dynamic prompt injection system that adjusts the output complexity based on the user's profile (**beginner**, **intermediate**, **expert**).
- **Citation State:** A dedicated store for formatted references to prevent re-generation errors.

Key Technical Capabilities

- **Dynamic Expertise Adaptation:** The agent adjusts its writing style based on the user's profile.
 - *Beginner:* Simple analogies, no jargon
 - *Intermediate:* Moderate technical detail (default).
 - *Expert:* High technical depth, focusing on limitations and conflicting evidence.
- **Citation Remapping & Integrity:** A unique feature of Agent A2 is its **Citation Remapping Protocol**. The AI initially cites evidence using array indices (e.g., **[0]**, **[5]**). Before showing the answer to the user, a logic layer automatically remaps these to sequential numbers (e.g., **[1]**, **[2]**) and verifies that every citation actually exists in the bibliography.

3.4.3 Data Visualization and Graph-Based Representation

The visualization module provides a simple and interactive way to explore scientific knowledge extracted from research papers and citations. Instead of manually reading many separate documents, users can view the information as a connected network. This makes it easier to understand how papers, methods, results, and research topics are related. Hidden patterns, dependencies, and trends become visible through clear graphical representations.

The system represents all scientific knowledge as a graph made up of nodes and edges. This approach is independent of any specific subject area and can be applied to engineering, medicine, social sciences, or any other research domain. Because the method relies only on general relationships between entities, it remains flexible and reusable across disciplines.

Nodes are the basic elements of the graph. A node may represent a research paper, an author, a method, a result, a dataset, or a topic. Every node has a unique identifier, a defined type, and descriptive information that explains what it represents. Nodes also contain an importance score that reflects their role in the overall network. Additional details such as publication date, section references, or links to original documents are stored to ensure that all information can be traced back to reliable sources.



Edges represent relationships between nodes. An edge connects two nodes and shows how they are linked. For example, an edge may indicate that one paper cites another, that a method is used to produce a result, or that two topics are closely related. Relationships may be directional, such as citations, or non-directional, such as conceptual similarity. Each edge stores the type of relationship, its strength, a measure of confidence, and time-related information.

All relationships are stored in two complementary forms. The adjacency matrix organizes connections in a structured table that is useful for analysis and calculations. The edge list records relationships individually and is better suited for interactive visualization and filtering.

To help users focus on the most relevant information, the system ranks nodes using several measures. Nodes with many connections are treated as important, since they often represent influential papers or widely used methods. Other measures identify nodes that act as bridges between different research areas or that are connected to other highly important nodes. The quality and reliability of relationships are also considered so that well-supported knowledge receives greater emphasis.

The system supports multiple types of relationships at the same time, including citations, method usage, result derivation, comparisons, and topical similarity. By combining explicit links with content-based connections, the graph provides a rich and complete view of the research landscape.

All visualizations follow a common pipeline. First, nodes and edges are extracted from scientific documents. The graph is then stored and analyzed to compute importance scores and to remove weak or unreliable links. Finally, the prepared graph can be displayed using different visualization styles.

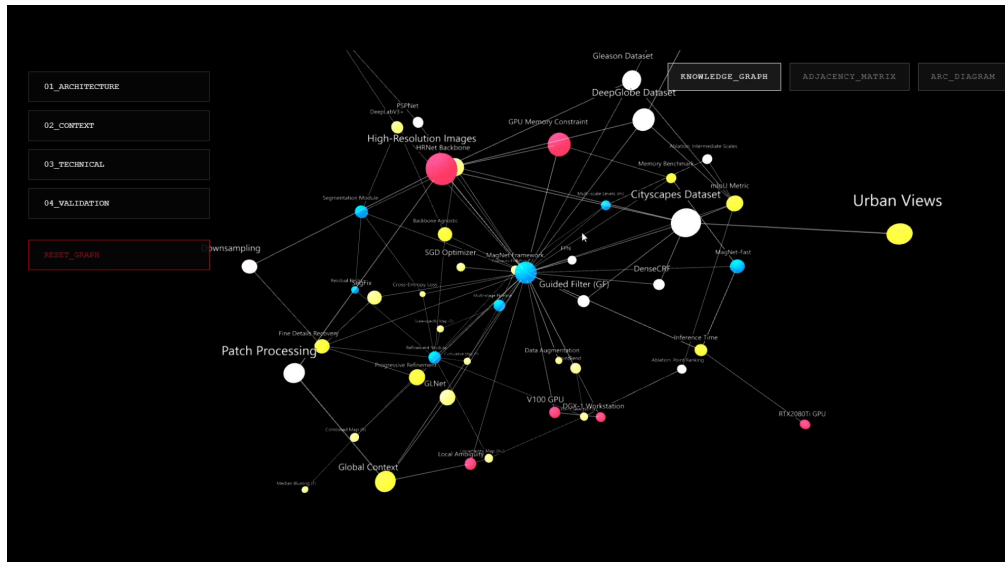
Force-directed layouts arrange nodes automatically so that related items appear close together, revealing clusters and central papers. Arc diagrams place nodes along a line and draw curved connections between them, making it easy to study citation flow and research evolution over time. Adjacency matrix views display the network as a grid, allowing users to quickly detect strong connections, dense clusters, and missing links.

These visual tools provide many benefits for research exploration. Users can identify important papers and methods, examine groups of related studies, and trace how ideas spread across publications. By expanding the neighborhood around any topic, researchers can follow direct and indirect influences. Areas with few connections highlight research gaps and potential opportunities for new work.

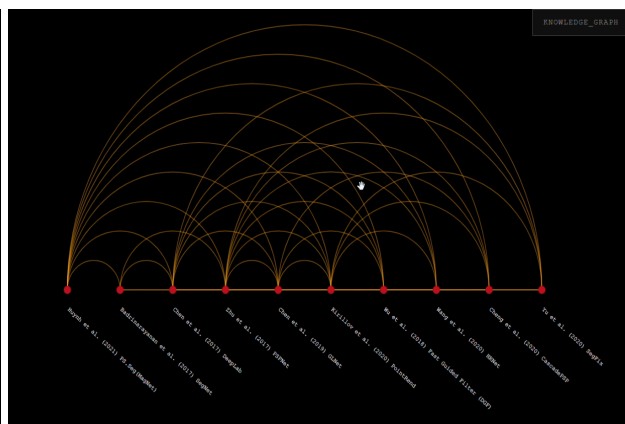
Because the entire framework relies only on general graph concepts, it remains fully domain-agnostic. The same techniques can be used to analyze any collection of scientific literature. Together, these visualization methods transform large sets of research documents into an organized and understandable knowledge environment, supporting faster discovery and better scientific understanding.

3.4.3.1 Algorithms Behind Visualization

The visualization module uses practical algorithms to convert raw graph data into clear and understandable visual forms. The main goal of these algorithms is not complex mathematics, but to present relationships between papers, methods, and concepts in a way that users can easily explore. The system supports three main visualization approaches: force-directed layouts, adjacency matrices, and arc diagrams. All three methods use the same underlying graph information but display it in different ways.



Force-directed layouts present the graph as if it were a physical system. Nodes are treated as objects that repel each other, while connections between nodes act like springs that pull related items together. The algorithm starts by placing nodes randomly and then repeatedly adjusts their positions based on these virtual forces. After many small adjustments, the layout becomes stable. The result is a natural arrangement where related papers and topics form visible clusters, and important, highly connected nodes appear near the center. This visualization is very useful for exploring ego networks, where a user can select one paper or topic and expand its neighborhood to discover related work.



Adjacency matrices represent the same network in a structured table form. All nodes are listed along the rows and columns, and each cell shows whether two nodes are connected and how strong

that connection is. This representation is especially useful for analysis because patterns in the matrix directly reveal influential nodes, tightly connected communities, and indirect relationships. It allows the system to rank important papers, detect research subfields, and compute paths between topics. Sparse areas in the matrix clearly show weakly explored regions, helping to identify potential research gaps.

Arc diagrams display the graph in a simple linear arrangement. Nodes are placed along a horizontal line, often ordered by time or importance, and relationships are drawn as curved arcs between them. This method is particularly effective for showing how ideas and citations flow over time. It allows users to visually follow chains of influence from older to newer papers and to recognize long-term trends. Missing or weak connections between parts of the diagram can highlight unexplored links between research areas.

Each of these visualization techniques reveals different aspects of the same knowledge graph. Force-directed layouts provide an intuitive overview of the entire research field. Adjacency matrices offer detailed analytical insight into connectivity and structure. Arc diagrams emphasize order and evolution. When used together, they enable researchers to identify important contributions, explore related studies, follow citation paths, and detect gaps in existing knowledge.

Because all methods rely only on general concepts of nodes and edges, the approach remains completely domain independent. The same algorithms and visual tools can be applied to literature from any scientific field. Through these techniques, the system transforms large collections of complex research documents into clear and interactive maps of knowledge, supporting better understanding and faster discovery.

See the link of the Knowledge Graph: [🔴 Challenge_Final.mp4](#)

3.4.4 Global Memory

The system includes a Redis-backed global memory layer, which stores session-related information and feedback. Examples of stored data include:

- Metadata about the main paper
- Execution summaries and reasoning context
- Feedback signals such as rejected or unreliable sources

Global memory is accessed only through A0, preventing unintended interactions between agents and ensuring consistency across system runs.

3.4.5 Execution Flow

The execution of a query follows a structured sequence:

1. The user submits a query along with a paper reference.
2. A0 performs policy planning and determines allowed operations.
3. A0 invokes A1 to ingest the paper and optionally retrieve external evidence.
4. Retrieved evidence is filtered according to policy constraints.
5. A0 invokes A2 to generate the final answer.
6. Execution metadata is stored in global memory for traceability and future feedback.

3.4.6 Design Advantages

This architecture offers several key advantages:

- Policy enforcement through a single point of control
- Traceability of evidence and decisions
- Modularity and extensibility, allowing new agents to be added easily
- Reduced hallucination risk by separating retrieval and reasoning
- Improved reliability through controlled memory access

3.5. ADDITIONAL DESCRIPTION OF POSSIBLE PARTICULARLY INNOVATIVE ASPECTS OF THE SOLUTION

The proposed solution introduces several particularly innovative aspects that distinguish it from existing tools for scientific literature exploration and AI-assisted research. These innovations are not limited to individual features, but emerge from the integrated design of the system, which combines transparency, structured reasoning, and user adaptability within a single platform, as demonstrated by the final prototype.

A key innovative aspect is the **multi-agent approach to scientific reasoning**. Instead of relying on a single AI model to perform all tasks, the system coordinates multiple specialised components



responsible for document analysis, citation extraction, reasoning, and explanation generation. This architecture allows the solution to handle complex scientific questions in a structured and modular way, improving robustness and interpretability. By decomposing the reasoning process into explicit steps, the system aligns with emerging best practices in trustworthy AI and supports more reliable outcomes when analysing complex or interdisciplinary research topics.

Another particularly innovative feature is the strong emphasis on **citation-aware and evidence-grounded answers**. Every response generated by the system is explicitly linked to the original scientific sources used to support it. Users can directly access the cited passages, explore references, and understand the context in which information is presented. This level of transparency addresses a critical limitation of many existing AI-based tools, which often provide convincing answers without clearly indicating their origin. By design, the solution prioritises verifiability and accountability, making it suitable for academic and professional environments where trust in information is essential.

The solution also introduces innovation through its **adaptive interaction based on user expertise level**. By allowing users to select an expertise level, the system dynamically adjusts the depth of explanations, the technical language used, and the amount of contextual detail provided. This feature enables the same platform to serve diverse user profiles, from students approaching a topic for the first time to expert researchers requiring concise and precise analysis. Such adaptability enhances usability and learning outcomes while reducing the need for multiple specialised tools.

A further innovative element is the **interactive refinement of answers through user feedback**. The prototype allows users to request shorter, more detailed, or reformulated answers, enabling an iterative dialogue between the user and the system. This human-in-the-loop mechanism transforms the AI from a static answer generator into a collaborative research assistant, supporting deeper understanding and continuous improvement of the output.

Finally, the system's design supports **structured exploration of scientific knowledge**, enabling users not only to ask questions about individual papers, but also to navigate citations and related works in a coherent way. This approach shifts the focus from isolated document analysis to a broader understanding of how scientific knowledge is connected and evolves over time.

Overall, the innovative aspects of the proposed solution lie in its **combination of multi-agent reasoning, transparent citation grounding, adaptive explanations, and interactive user engagement**. Together, these elements contribute to a novel and trustworthy approach to AI-assisted scientific research, positioning the solution as a meaningful advancement over existing literature exploration tools.



3.6. DEVELOPMENT STATUS OF THE SOLUTION

(Please indicate the current stage of development and Potentially reachable by TRL coding - Technology Readiness Level)

At the end of the Challenge, the proposed solution has reached the stage of a **functional prototype**, demonstrating the core concepts, architecture, and user interactions envisioned during the design phase. The final prototype shows that the system is technically feasible and capable of addressing the identified problem, even though it is not yet a fully mature or industrial-grade product.

From a development perspective, RefHunters can be classified as a **proof-of-concept with validated functionalities**. The prototype includes a working web-based interface, the integration of a multi-agent AI architecture, and the ability to analyse scientific papers, answer user questions, and provide citation-backed explanations. Key features such as document upload, interactive question answering, citation tracing, reference exploration, and adaptive explanations based on user expertise level are implemented and demonstrable. This confirms that the underlying technical approach is sound and that the solution can operate in a realistic usage scenario.

In terms of **Technology Readiness Level (TRL)**, the current development status can be positioned at **TRL 4–5**.

- At **TRL 4**, the technology has been validated in a laboratory environment, with core components working together in a controlled setting.
- Moving toward **TRL 5**, the system has been tested in a relevant environment through realistic use cases, such as analysing real scientific papers and responding to complex user queries, as shown in the final prototype demonstration.

However, the solution has not yet reached higher TRL levels, as several aspects would need further development before real-world deployment. These include robustness testing at scale, performance optimisation, security hardening, and integration with external systems such as institutional repositories or enterprise knowledge platforms. Additionally, broader evaluation with real users and larger document collections would be required to assess scalability, usability, and long-term reliability.

Looking forward, with further development and investment, the solution could realistically progress to **TRL 6 or 7**. This would involve deploying the system in a pilot environment within an academic institution or corporate R&D setting, integrating additional features such as cross-paper analysis and visualisation tools, and refining the user interface based on user feedback. Achieving these levels would demonstrate the solution's readiness for operational use and early adoption.

In conclusion, the current development status reflects a **strong and coherent prototype**, successfully demonstrating the core innovative ideas of the project. While not yet market-ready, the

solution has reached a level of maturity that clearly shows its potential for further development, validation, and eventual deployment in academic and professional contexts.

4. SECTION: STATE OF THE ART

4.1. DESCRIPTION OF THE KNOWN SOLUTIONS PRESENT IN LITERATURE/ PAPERS

Paper Title	Authors	Brief description of the Paper	Differences with the Proposed Solution
RAG: Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks	Lewis et al.	Introduces retrieval-augmented generation combining neural retrieval with language models to answer knowledge-intensive questions.	Focuses on general knowledge tasks; does not provide fine-grained citation tracing or interactive exploration of scientific papers.
SciBERT: A Pretrained Language Model for Scientific Text	Beltagy et al.	Presents a domain-specific language model trained on scientific corpora to improve NLP tasks in academia.	Improves text understanding but does not address explainability, source validation, or cross-paper reasoning.
ScholarQ: Question Answering on Scholarly Documents	Yang et al.	Proposes QA systems specifically designed for scholarly documents using IR and NLP techniques.	Limited transparency on citation usage and lacks user-controlled expertise adaptation.
Explainable Question Answering over Scientific Literature	Thorne et al.	Explores explainability in scientific QA by linking answers to evidence sentences.	Focuses on explainability but does not support interactive exploration,

			expertise-level adaptation, or knowledge graph generation.
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4.2. PRIOR ART - EXISTING PATENTS

(Search on web site: www.orbit.com or Google Patent or www.espacenet.com)

Patent Number	Brief Patent description	Differences with the Proposed Solution
US20190251476A1	AI-based document analysis system for extracting and summarizing information from large text corpora.	Primarily oriented to enterprise document management; lacks scientific citation tracing and cross-paper exploration.
US20200372814A1	System for automated literature review using machine learning and semantic analysis.	Focuses on automation rather than transparency and user-driven exploration of sources.
EP3559063A1	Method for knowledge extraction and reasoning over document collections.	Does not include explicit support for explainable scientific QA or expertise-level adaptation.

4.3. KNOWN PRODUCT

(Description of any products that have similarity, same functioning and/or final objective, compared to the proposed solution)

Product Name	Manufacturing Company	Brief description and features of the Product	Differences with the Proposed Solution
Semantic Scholar	Allen Institute for AI	Academic search engine offering paper	Provides discovery but limited

		discovery, citation graphs, and AI-generated summaries.	interactive QA and no adaptive explanations based on expertise level.
Elicit	Ought	AI assistant for literature review, summarizing papers and extracting key findings.	Answers are often abstract-level and lack fine-grained citation traceability within documents.
Scite.ai	Scite Inc.	Citation analysis platform classifying citations as supporting or contrasting.	Focuses on citation context but does not provide interactive question answering or paper-level exploration.
ChatGPT	OpenAI	General AI model capable of summarizing and answering questions on scientific topics.	Not restricted to verified sources and lacks explicit source traceability and document-level transparency.

The analysis considers representative academic works, patents, and commercial products addressing scientific information retrieval and analysis. The proposed solution differentiates itself by combining verified-source reasoning, citation transparency, expertise-level adaptation, and interactive exploration within a single integrated system.

5. SECTION: BUSINESS

5.1.DEFINITION OF VALUE PROPOSITION

The value proposition of the proposed solution is to provide a reliable, transparent, and efficient way to access and explore scientific knowledge through AI-assisted reasoning grounded in verified sources. RefHunters addresses the limitations of traditional academic search tools and generic AI



systems by combining question answering, citation traceability, and structured exploration within a single platform.

The solution creates value by significantly reducing the time and effort required to analyse scientific literature, while increasing trust in AI-generated outputs. Every answer is explicitly linked to its original sources, allowing users to verify information, explore citations, and understand the context in which knowledge is produced. This feature is particularly valuable in academic, professional, and regulated environments, where accuracy and explainability are essential.

In addition, RefHunters offers adaptive explanations based on the user's expertise level, making scientific content accessible to beginners while still supporting in-depth analysis for expert users. This flexibility broadens the potential user base and enhances usability across different domains.

Overall, the value proposition lies in enabling faster, more informed, and evidence-based decision-making, improving research productivity, reducing cognitive load, and supporting transparent knowledge management in both academia and industry.

5.2.DEFINITION OF ONE, OR MORE, “USE CASE” FOR THE USE OF THE SOLUTION

The proposed solution is designed to support different use cases related to the exploration, analysis, and validation of scientific knowledge. Based on the final prototype and the demonstrated slides, one main representative use case is described below.

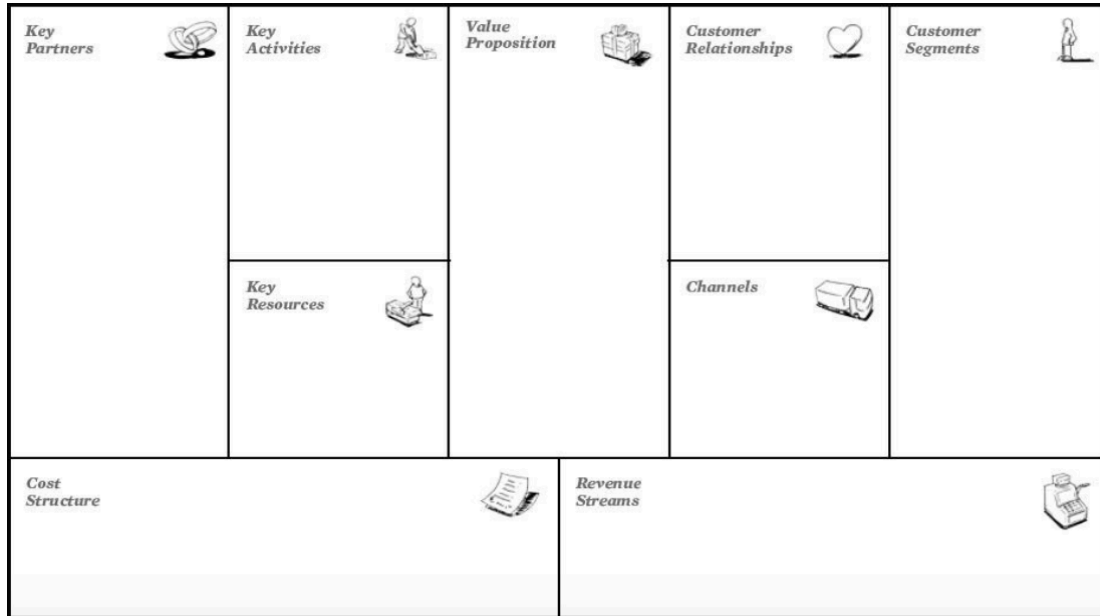
A primary use case concerns scientific literature analysis and comparison. In this scenario, a user uploads a scientific paper in PDF format and selects an expertise level (Beginner, Intermediate, or Expert). The system analyzes the document and allows the user to ask questions such as summaries, methodological explanations, or comparisons with related works. Answers are generated using only verified scientific sources and are always accompanied by explicit citations, enabling transparency and trust. Through iterative feedback, the user can request more detailed or shorter answers, adapting the output to their specific needs.

A second use case focuses on comparative research and critical evaluation. The user asks cross-paper questions, for example, comparing a proposed method with previous approaches in the literature. By adjusting the expertise level, the same question can be explored from different perspectives, supporting both learning-oriented users and expert researchers. The integration of citation tracing and, in future developments, knowledge graph visualization further supports the understanding of relationships between papers, authors, and research trends.

Overall, these use cases highlight how RefHunters can be applied in academic and professional contexts to improve literature review processes, support evidence-based analysis, and reduce the time and effort required to extract reliable insights from scientific documents.

5.3.BUSINESS MODEL CANVAS

(Scheme filling and a short description of the 9 modules)



- 1- **Key Partners:** academic publishers and open-access repositories providing publicly accessible scientific papers, as well as research institutions and universities supporting validation and testing. Additional partners may include cloud service providers for infrastructure and AI technology providers for language models and NLP components.
- 2- **Key Activities:** development and maintenance of the multi-agent AI system, scientific document processing, citation extraction, and answer generation. Continuous improvement of reasoning quality, transparency mechanisms, and user interface design are also essential activities.
- 3- **Key Resource:** AI models and multi-agent architecture, the software platform, access to scientific repositories, and technical expertise in AI, natural language processing, and knowledge management.
- 4- **Value Proposition:** transparent, citation-backed AI assistance for scientific literature analysis. It reduces research time, increases trust in AI-generated answers, and supports users with adaptive explanations based on expertise level, enabling evidence-based decision-making in academic and professional contexts.
- 5- **Customer Relationship:** self-service and interactive usage, supported by clear feedback mechanisms that allow users to improve answer quality. For institutional clients, relationships may include technical support, onboarding, and customization services.
- 6- **Channels:** the solution is delivered primarily through a web-based platform. Additional channels include institutional deployments, integrations with digital knowledge management systems, and direct collaboration with universities and research organizations.

- 7- **Customer Segments:** students and researchers, academic institutions, corporate R&D and innovation units, consulting and advisory firms, and knowledge-intensive industries such as healthcare, engineering, and high-tech sectors.
- 8- **Cost Structure:** mainly includes software development, cloud infrastructure, AI model usage, and system maintenance costs. Additional costs may arise from data processing, scaling, and customer support for enterprise users.
- 9- **Revenue:** subscription-based access for institutions and enterprises, custom deployments for regulated industries, and premium features such as advanced analytics, comparative reasoning, and visualization tools.

5.4.S.W.O.T. ANALYSIS

Related to the proposed solution, indicate with a brief description:

5.4.1. Strengths

A major strength of the proposed solution is its **high level of transparency and explainability**. Unlike generic AI tools, RefHunters generates answers that are always supported by explicit citations, allowing users to trace information back to its original scientific sources. This feature significantly increases trust and usability in academic and professional contexts.

Another strength is the **adaptive interaction based on expertise level**, which enables the same system to support beginners, students, and expert researchers without changing tools. **The multi-agent architecture** further strengthens the solution by enabling structured reasoning, citation-aware analysis, and interactive exploration of scientific papers, reducing both research time and cognitive effort.

5.4.2. Weaknesses

One of the main weaknesses of the current prototype is its **dependence on publicly accessible scientific sources**, which limits the analysis of paywalled or proprietary publications. This may reduce completeness in some research domains.

In addition, the system's performance depends on the **accuracy of document parsing and citation extraction tools**, which can sometimes lead to partial or noisy results, especially for complex PDFs. As a prototype developed within a Challenge context, the solution is also not **yet optimized for large-scale deployment** in terms of computational efficiency, response time, and integration with existing enterprise systems.

5.4.3. Opportunities

The solution addresses a rapidly growing need for AI-assisted, trustworthy research tools in academia, corporate R&D, consulting, and regulated industries. Increasing regulatory attention

on AI transparency, explainability, and auditability represents a strong opportunity, as RefHunters is already aligned with these requirements.

Furthermore, planned future developments (such as cross-paper comparative analysis, tracking the evolution of ideas, and citation network visualization) can significantly differentiate the solution from existing tools. These features open opportunities for institutional adoption, partnerships with universities, and potential commercialization through enterprise subscriptions or tailored deployments.

5.4.4. Threats

A key threat is competition from **large technology companies and established academic platforms** that may integrate similar features into existing products. **Changes in copyright regulations or access policies of scientific publishers** could also limit data availability and system functionality.

Finally, the fast pace of AI innovation represents a risk, as maintaining competitiveness will require **continuous model updates, technical maintenance, and regulatory compliance**, which may increase long-term costs and development complexity.

5.5. COMPETITORS

(Description of any possible and/or potential Competitors)

The proposed solution operates in a competitive landscape that includes academic search engines, AI-assisted research tools, and general-purpose language models. While several existing solutions partially address the problem of scientific information retrieval and analysis, none fully combine transparency, citation traceability, adaptive reasoning, and interactive exploration in the way demonstrated by the final RefHunters prototype.

A first group of competitors is represented by academic search platforms, such as Semantic Scholar and Google Scholar. These tools are highly effective for discovering scientific papers, tracking citations, and identifying influential works. However, they mainly provide lists of documents and metadata, leaving the user responsible for reading, interpreting, and synthesising information across multiple papers. In contrast, RefHunters goes beyond discovery by enabling question answering directly on the paper content, supported by explicit citations and contextual explanations, thus reducing manual effort and cognitive load.

A second category includes AI-powered literature review assistants, such as Elicit or similar tools. These systems offer summarisation and extraction of key information from scientific articles, helping users speed up literature reviews. However, their answers are often limited to abstracts or high-level summaries and may lack fine-grained traceability to specific sections or passages. Moreover, they usually do not allow users to interactively refine answers or adapt explanations



based on expertise level. RefHunters differentiates itself by providing citation-aware answers at passage level, user feedback mechanisms (e.g. “more detailed” or “shorter”), and adaptive reasoning tailored to beginner or expert users.

Another important competitive reference is general-purpose AI language models, such as ChatGPT when used for scientific questions. These models are flexible and powerful in generating natural language responses, but they are not restricted to verified scientific sources and often do not clearly indicate where information comes from. This limits their reliability and suitability for academic or regulated contexts. RefHunters addresses this limitation by relying exclusively on verified scientific sources, explicitly communicating which parts of the literature have been analysed and linking every answer to its original references.

Finally, some platforms focus on citation analysis and research evaluation, such as Scite.ai, which classifies citations according to their role (supporting, contrasting, or mentioning). While these tools provide valuable insights into citation context, they do not support interactive question answering, cross-paper comparisons driven by user queries, or adaptive explanations. RefHunters integrates citation analysis into a broader reasoning framework, allowing users to both explore and question the scientific literature in a unified workflow.

Overall, while existing competitors address specific aspects of scientific knowledge exploration (such as discovery, summarisation, or citation analysis) the proposed solution differentiates itself by offering an integrated, transparent, and user-adaptive system. By combining AI-assisted reasoning, explicit citation traceability, expertise-level adaptation, and interactive exploration, RefHunters positions itself as a complementary and, in many use cases, more advanced alternative to current solutions in academia, research-intensive industries, and consulting environments.

6. SECTION: FUTURE DEVELOPMENTS

6.1. POSSIBLE FUTURE DEVELOPMENTS

(Describe any steps for the future developments of the Solution)

Building on the current prototype, several future developments are planned to further strengthen the analytical capabilities and usability of the solution.

A first improvement concerns the **automatic generation of tables and figures** from scientific papers. This feature would support users in synthesising key results, comparisons, and experimental outcomes, significantly reducing the time required to prepare literature reviews and technical reports.

Another important direction is the introduction of multi-LLM, citation-aware reasoning. This **advanced reasoning** capability would allow the system to answer complex research questions



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faster and by combining multiple LLM's models for reasoning, it'll guarantee the best results in the fastest time possible.

The system can also be extended to support **cross-paper comparative analysis**, enabling users to directly compare methods, assumptions, and results across different scientific works. This would facilitate critical evaluation of competing approaches and support more informed decision-making.

Closely related to this, future developments include **tracking the evolution of ideas across related works**. By analysing citations and thematic links over time, the system could help users understand how concepts and methodologies develop within a research field and make it easier to identify research gaps.

Additionally, the integration of **citation and author network visualisation** would provide an interactive graphical representation of relationships between papers, authors, and research communities, helping users identify influential works and key contributors.

Finally, **automated structuring of scientific literature** would further improve usability by organising papers into coherent thematic sections, supporting faster navigation and more effective knowledge exploration.

DATA PROCESSING

Data processing is authorized according to General Data Protection Regulation (EU) 2016/679 (GDPR) for the purposes contained in this application.